

# Causal Audit–Return Duality for Streaming Quantum Memory

An exact classical-memory frontier and a coherent one-qubit separation

Causal Audit–Return Memory Project  
Open research artifact, version 1.1

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## Abstract

We introduce a two-branch benchmark for coherent memory in a sequential quantum process. A trusted verifier sends one half of each of  $n$  independent EPR pairs through a device, requires every output to be committed before the next input is released, and sequesters those outputs. After the stream and the device’s terminal-memory commitment, a hidden choice requests either (AUDIT) the parity of later computational-basis measurements on the references or (RETURN) joint recovery of all EPR pairs and visible reset of the terminal memory. For the explicit null class of arbitrary adaptive local instruments with unlimited classical memory and disposable within-slot ancillas, but no coherent state or entanglement persisting between slots, we prove the exact support

$$C_{n,\lambda}^{\text{stream}} = \max_{0 \leq t \leq 1} \left[ \frac{\lambda}{2}(1+t^n) + (1-\lambda) \left( \frac{1+\sqrt{1-t^2}}{2} \right)^n \right].$$

The reduction covers non-QND instruments and arbitrary transcript-conditioned joint recovery. At equal branch weight, the null is  $3/4$  for every  $n \geq 2$ , whereas one persistent coherent qubit attains  $(P_A, F_R) = (1, 1)$ . A collective instrument retaining only a classical outcome has the strictly larger support  $C_\lambda^{\text{coll}} = \frac{1}{2} + \frac{1}{2}\sqrt{\lambda^2 + (1-\lambda)^2}$ , isolating the causal cost of streaming. For  $n \geq 3$ , the optimal classical strategy exhibits a first-order transition from recording nothing to an almost projective per-slot measurement. We supply a fixed-sample statistical certificate with explicit systematic allowances, conservative power planning, adversarial numerical checks, deterministic data and figures, and a hardware-facing protocol. The result is a trusted-interface temporal quantum-memory witness, not a device-independent claim, a proof of global erasure, or evidence for a preferred interpretation of quantum mechanics.

**Keywords:** quantum memory; multitime quantum processes; information–disturbance; entanglement fidelity; reversible computation; causal benchmarks.

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# 1 Introduction

A memory is operationally quantum only when coherence carried across time changes what a process can do. Merely observing a reversible circuit, a final interference fringe, or an auxiliary register returned to zero is not enough: a known closed circuit can be compiled into an endpoint-equivalent channel that never instantiates the advertised intermediate record. The scientific question therefore needs an access model and a comparison class, not interpretive vocabulary.

We study a streamed task with two incompatible late uses of one committed process. The verifier supplies fresh entangled carriers one at a time and permanently removes each returned carrier from the device. Only after the final slot does a hidden choice ask the committed memory either to reveal a temporal predicate or to help reverse the disturbance created while learning it. The predicate is the parity of  $n$  later reference outcomes. The reversal target is global entanglement fidelity of all  $n$  returned carriers. These endpoints force the same prefix to be useful as a record and reversible as a coherent process.

The ingredients are familiar. Information acquisition competes with disturbance [1]; read-versus-return tasks appear in quantum sealing [2]; late path-versus-phase games quantify operational duality [3]; and parity discrimination has a long history in quantum cryptography [4]. Multitime process formalisms distinguish quantum, classical, and memoryless transmission across slots [5, 6, 7]. Sequential random-access tasks and bounded-memory channel discrimination already witness quantum memory under other scores and assumptions [8, 9]. Our claimed contribution is not any of these ingredients alone.

For a precisely declared classical-memory comb, we derive the complete audit–return support function. The proof first removes arbitrary non-QND output dynamics through a flagged entanglement-fidelity bound. Causal factorisation then turns both the posterior parity bias and the maximum return fidelity into products along every refined classical path. Adaptation becomes an average of deterministic strength profiles, and a convexity argument shows that equal local strengths attain the optimum. This supplies an exact causal tensorisation rather than a symmetric ansatz.

Three consequences make the result experimentally legible. First, at the balanced point the unlimited adaptive classical-memory score is  $3/4$  for every stream length  $n \geq 2$ . Second, allowing the carriers to be processed collectively before producing a classical record raises the score to  $(1 + 1/\sqrt{2})/2$ , so sequestration is a physical resource restriction rather than cosmetic timing. Third, one coherent accumulator qubit reaches unit score and visibly resets in RETURN. The benchmark therefore compares three resource classes on the same task:

$$\text{streaming classical memory} < \text{collective classical record} < \text{one coherent temporal qubit}$$

for every nontrivial support direction, with the final inequality understood at the algebraic point.

We accompany the theorem with a fixed-sample statistical rule, a preregistration template, mandatory shortcut controls, deterministic frontier data, and a replaceable phenomenological error model. The goal is a benchmark that can be falsified. A positive result rejects the declared classical-memory interface. It does not identify a unique microscopic implementation, certify an unrestricted hidden device, prove deletion of inaccessible environmental records, or support claims about observers or alternate worlds.

## 1.1 Contributions

The paper provides:

1. an operational AUDIT/RETURN task with immediate output sequestration and a terminal-memory commitment;

2. an exact theorem for arbitrary adaptive, non-QND, classical-memory streaming instruments;
3. closed forms, a first-order strategy transition for  $n \geq 3$ , and a strict collective-classical comparator;
4. an explicit one-qubit coherent strategy attaining perfect audit and return;
5. a conservative finite-statistics certificate with systematic-error and null-model slack;
6. reproducible analytic and numerical validation, planning data, controls, and implementation guidance.

All novelty language is qualified by the dated literature audit. The safe claim is the exact streamed parity-audit versus global-entanglement-return frontier and its resource separation, to our knowledge—not weak measurement, parity processing, delayed choice, reversible computation, or quantum-memory witnessing in general.

## 2 Scope and related work

### 2.1 The comparison class is the claim

Quantum memory has several inequivalent meanings in a multitime process. Process tensors and quantum combs provide the general slot structure [5, 6]. Memory cost can be a global property of a strategy even when classical assistance is free [10]. Recent classifications distinguish processes that transmit coherent quantum systems, classical information through conditional instruments, or no information at all [7]. We use a direct operational restriction: after refining every outcome and shared-random seed, each leaf channel factorises over slots. Disposable quantum ancillas are allowed inside a slot; no coherent state or entangled common cause persists between slots.

This null is stronger than saying that some reduced channel is entanglement breaking and weaker than demanding memorylessness. Unlimited classical state, adaptive choice, arbitrary finite alphabets, non-QND maps, and transcript-conditioned joint recovery are included. A positive experiment is therefore a resource witness only relative to trusted timing, source, sequestration, and terminal-port assumptions.

### 2.2 Information–disturbance and operational duality

One-shot estimation–operation fidelity tradeoffs establish that extracting classical information disturbs quantum states [1]. Operational wave–particle duality similarly frames a late game between path and phase information [3]. Quantum seals study how readability limits later verification of an undisturbed state [2]. Our one-slot curve is not new: it is the binary information–return relation. The new-looking step is causal tensorisation for fresh sequestered inputs, a parity audit from the retained temporal record, and a global all-carrier return score under the full adaptive classical-memory null.

Parity itself is also occupied. Collective measurements can reveal global parity information unavailable to separate measurements in quantum cryptographic ensembles [4]. Here parity is chosen because a coherent accumulator needs only one qubit while any classical streamed strategy multiplies local posterior biases. The task is not cryptographic and does not claim secrecy.

### 2.3 Temporal-memory witnesses

Memory witnesses based on multitime process entanglement, effective environment dimension, and identity-channel behaviour already exist [11, 12, 13]. Sequential random-access codes can separate classical and quantum temporal memory under semi-device-independent assumptions

[8]. Bounded coherent and classical memory in adaptive discrimination can be formulated through constrained separability [9]. Our score differs because its second branch asks for joint entanglement return of the same committed carriers, rather than a later decoding success. This is an adjacent benchmark, not a replacement for those frameworks.

## 2.4 Reversible computation and interpretation

The honest strategy uses CNOT accumulation and uncomputation, established reversible-computing operations [14]. Its value comes from the late branch and access constraints, not from describing the circuit as multiple histories. The observed probabilities are fully predicted by standard quantum mechanics. No interpretation-specific claim follows.

## 2.5 Priority discipline

An adversarial search through 12 August 2026 found no primary source stating the full theorem with all of: fresh independent entangled inputs; immediate output sequestration; arbitrary adaptive non-QND instruments with classical but no coherent inter-slot memory; late transcript-only parity audit; joint all-carrier entanglement return; and the same exact support and phase transition. Literature searches cannot prove absence. The priority claim should be withdrawn if an earlier theorem specialises directly to this interface and score.

Table 1: Nearest occupied territory and the remaining distinction.

| Area                      | Established result                                | Distinction here                                                    |
|---------------------------|---------------------------------------------------|---------------------------------------------------------------------|
| Information–disturbance   | Tight one-shot fidelity tradeoffs                 | Exact causal product law for a parity audit and global return       |
| Ways/Phases games         | Late operational duality choices                  | Fresh time slots, sequestered outputs, and retained temporal memory |
| Parity discrimination     | Collective advantage for global parity            | Same-prefix reversal score and a classical-memory comb null         |
| Quantum seals             | Readability versus later verification             | Multitime stream and all-carrier EPR return                         |
| Temporal memory witnesses | Classical/quantum hierarchy and sequential scores | Exact same-task support plus coherent one-qubit attainment          |
| Reversible computing      | Compute–uncompute circuits                        | Hidden late challenge and a falsifiable resource separation         |

## 3 Operational model

Fix  $n \geq 1$ . At slot  $i$ , a trusted verifier prepares

$$|\Phi\rangle_{R_i A_i} = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

sends  $A_i$  to the device, receives  $B_i$ , and sequesters  $B_i$  before releasing  $A_{i+1}$ . After the last output, the device commits a terminal memory port  $M$  to the verifier. A hidden coin then selects one branch.

**AUDIT.** The verifier measures every  $R_i$  in  $Z$ , obtaining independent uniform bits  $X_i$ , and decodes from  $M$  a guess for

$$Z = X_1 \oplus \cdots \oplus X_n.$$

The sequestered  $B_i$  are unavailable. Let  $P_A$  be the success probability.

**RETURN.** The verifier applies a fixed joint decoder to  $MB_1 \cdots B_n$  and accepts with

$$\Pi_R = |0\rangle\langle 0|_M \otimes \bigotimes_{i=1}^n |\Phi\rangle\langle \Phi|_{R_i B_i}.$$

Let  $F_R$  be the acceptance probability. The visible reset  $M = 0$  does not imply deletion of any inaccessible purification of a classical record.

For  $0 \leq \lambda \leq 1$ , define the support score

$$S_{n,\lambda} = \lambda P_A + (1 - \lambda) F_R.$$

The balanced benchmark uses  $\lambda = 1/2$ .

### 3.1 Adaptive classical-memory streaming null

The null may use arbitrary finite classical memory and shared randomness. In slot  $i$ , conditional on the earlier classical history  $h = k_{<i}$ , it applies an arbitrary instrument

$$\{\mathcal{N}_{k_i}^{(i,h)} : A_i \rightarrow B_i\}_{k_i}.$$

It may create and discard arbitrary within-slot quantum ancillas. It may not retain a coherent system across the slot boundary, begin with entanglement shared between slot ancillas, revisit a committed  $B_i$ , or receive a  $B_i$  in AUDIT. In RETURN, the full classical transcript may control an arbitrary joint recovery on all sequestered outputs.

After refining all classical random seeds and outcomes, every leaf transcript  $k = (k_1, \dots, k_n)$  has a conditional channel that factorises over slots. For computational-basis inputs  $x = (x_1, \dots, x_n)$ , its likelihood is

$$p(k|x) = \prod_{i=1}^n p_i(k_i|x_i, k_{<i}). \quad (1)$$

This factorisation is the mathematical null. The experiment must support it through timing and isolation evidence; the score alone cannot enforce a laboratory boundary.

### 3.2 Local audit strength and return curve

At a realised prefix  $h$ , define

$$r_i(k_i|h) = \frac{p_i(k_i|0, h) + p_i(k_i|1, h)}{2}$$

and, when  $r_i > 0$ , define a signed posterior bias by

$$s_i \delta_i = \frac{p_i(k_i|0, h) - p_i(k_i|1, h)}{p_i(k_i|0, h) + p_i(k_i|1, h)}, \quad 0 \leq \delta_i \leq 1. \quad (2)$$

The one-slot return curve is

$$f(t) = \frac{1 + \sqrt{1 - t^2}}{2}. \quad (3)$$

Here  $t = 0$  extracts no computational-basis information and preserves an EPR pair perfectly;  $t = 1$  permits a perfect bit record and leaves entanglement fidelity  $1/2$ .

### 3.3 A saturating null family

For any  $t_i \in [0, 1]$ , the binary diagonal instrument

$$K_0^{(i)} = \begin{pmatrix} \sqrt{(1+t_i)/2} & 0 \\ 0 & \sqrt{(1-t_i)/2} \end{pmatrix}, \quad K_1^{(i)} = \begin{pmatrix} \sqrt{(1-t_i)/2} & 0 \\ 0 & \sqrt{(1+t_i)/2} \end{pmatrix}$$

produces local audit bias  $t_i$  and return fidelity  $f(t_i)$ . The transcript parity is the optimal AUDIT guess and identity recovery saturates the RETURN expression derived below.

## 4 Exact frontiers and resource separation

**Theorem 4.1** (Streaming parity audit–return frontier). *For the adaptive classical-memory null in Section 3, including arbitrary non-QND local instruments and arbitrary transcript-conditioned joint recovery,*

$$C_{n,\lambda}^{\text{stream}} = \max_{0 \leq t_1, \dots, t_n \leq 1} \left[ \frac{\lambda}{2} \left( 1 + \prod_{i=1}^n t_i \right) + (1 - \lambda) \prod_{i=1}^n f(t_i) \right] \quad (4)$$

$$= \max_{0 \leq t \leq 1} \left[ \frac{\lambda}{2} (1 + t^n) + (1 - \lambda) f(t)^n \right]. \quad (5)$$

The bound is attainable by independent weak diagonal instruments.

The complete proof is in Appendix A. Its logical sequence is worth stating. For each transcript, a Choi-positivity lemma upper-bounds recoverable EPR fidelity using only computational-basis likelihoods. Equation (1) then makes the likelihood a product. Bayes' rule makes the posterior parity bias a product of  $\delta_i$ , while the recovery lemma makes EPR return a product of  $f(\delta_i)$ . An adaptive protocol is an average of realised strength profiles and cannot exceed the largest profile. Finally, the substitution  $t_i = 2y_i/(1 + y_i^2)$  and strict convexity of  $\log(1 + e^{2z})$  show that equal  $y_i$ , hence equal  $t_i$ , optimise every linear support.

**Corollary 4.2** (Balanced null). *For  $n \geq 2$  and  $\lambda = 1/2$ ,*

$$C_{n,\lambda}^{\text{stream}} = \frac{3}{4}.$$

The least-disturbing optimum is  $t = 0$ : the null learns no parity beyond chance and returns every EPR pair.

### 4.1 Closed forms and the strategy transition

For one slot,

$$C_{1,\lambda}^{\text{stream}} = \frac{1}{2} + \frac{1}{2} \sqrt{\lambda^2 + (1 - \lambda)^2}.$$

For two slots,

$$C_{2,\lambda}^{\text{stream}} = \begin{cases} 1 - \lambda/2, & 0 \leq \lambda \leq 1/2, \\ \lambda/2 + \lambda^2/(3\lambda - 1), & 1/2 < \lambda \leq 1. \end{cases} \quad (6)$$

The optimal strength grows continuously from zero at  $\lambda = 1/2$ .

For  $n \geq 3$ , the transition is first order. Let  $z_n$  be the unique solution

$$(1 - z_n)(1 + z_n)^{n-1} = 1, \text{quad} z_n \in \left( \frac{n-2}{n}, 1 \right). \quad (7)$$

Then

$$\lambda_c(n) = \frac{1}{1 + 2^{n-2} z_n^{(n-2)/2} (1 - z_n)}. \quad (8)$$

Below  $\lambda_c$ , the optimum is  $t = 0$ . At  $\lambda_c$ , it ties a finite-strength branch, so the optimal strength jumps discontinuously. Numerically,

$$(\lambda_c, t_c) = (0.6247789, 0.9717365), (0.6495455, 0.9961752), (0.6588955, 0.9992936)$$

for  $n = 3, 4, 5$ . Moreover  $\lambda_c(n) \rightarrow 2/3$  from below. The optimal null thus changes from recording nothing to recording each slot almost projectively rather than moving through a broad weak-measurement phase.

## 4.2 Collective classical-record comparator

Now remove streaming sequestration and allow one joint instrument on all carriers, while requiring that only a classical outcome survives the commit point.

**Proposition 4.3** (Collective classical record). *For every  $n$ ,*

$$C_\lambda^{\text{coll}} = \frac{1}{2} + \frac{1}{2} \sqrt{\lambda^2 + (1 - \lambda)^2}. \quad (9)$$

*For  $n \geq 2$  and  $0 < \lambda < 1$ , this is strictly larger than  $C_{n,\lambda}^{\text{stream}}$ .*

A weak measurement of the two parity subspaces attains (9). Optimality follows by grouping every outcome's likelihoods into even and odd sectors and applying the classical fidelity–total-variation circle; see [Appendix A](#). This comparator is not a coherent device: it retains only a classical outcome. It quantifies exactly what becomes possible if timing ceases to enforce causality.

## 4.3 One coherent qubit closes the gap

Initialise a memory qubit  $M = |0\rangle$ . In slot  $i$ , apply  $\text{CNOT}_{A_i \rightarrow M}$  and return  $A_i$ . After the final slot, commit  $M$ .

- In AUDIT, measuring  $M$  in  $Z$  gives the parity of the later  $Z$  measurements on the references.
- In RETURN, applying  $\text{CNOT}_{B_i \rightarrow M}$  for every sequestered carrier uncomputes  $M$  to  $|0\rangle$  and exactly restores all EPR pairs.

Thus  $(P_A, F_R) = (1, 1)$ . The operational resource is a coherent temporal accumulator, not an observer or an interpretation of quantum mechanics.



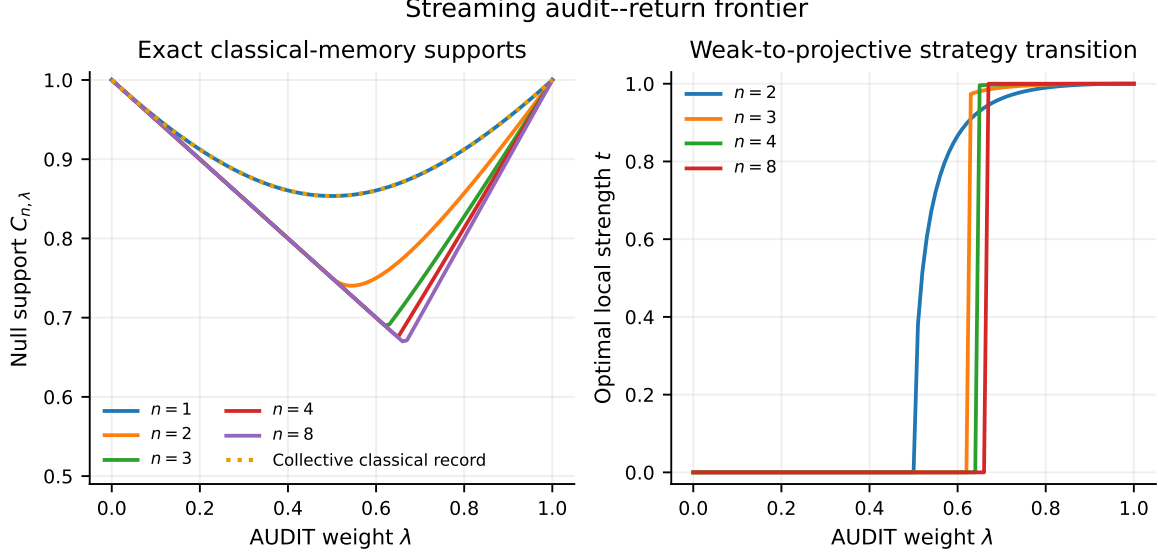


Figure 1: Exact streaming supports and exposed local strengths. At  $n \geq 3$ , the optimal classical strategy jumps from no record to an almost projective record as the audit weight crosses  $\lambda_c(n)$ . The dotted curve is the collective classical-record comparator, not an unrestricted coherent strategy.

## 5 Statistical certificate and experimental protocol

The theorem gives a probability bound. An experiment needs a fixed analysis rule that accounts for finite samples and declared model uncertainty.

### 5.1 Fixed-sample certificate

Let  $(K_A, N_A)$  and  $(K_R, N_R)$  be successes and fixed trials in the two branches. Define

$$\hat{S} = \lambda \frac{K_A}{N_A} + (1 - \lambda) \frac{K_R}{N_R}.$$

For independent Bernoulli trials, the one-sided weighted Hoeffding radius is

$$r_\alpha = \sqrt{\frac{1}{2} \left( \frac{\lambda^2}{N_A} + \frac{(1 - \lambda)^2}{N_R} \right) \log \frac{1}{\alpha}}. \quad (10)$$

Before observing outcomes, let  $\delta_A$  and  $\delta_R$  upper-bound possible upward bias in the two estimators, and let  $\delta_{\text{null}}$  enlarge the theoretical null for source, isolation, and interface uncertainty. We certify incompatibility with the declared classical-memory null only if

$$\hat{S} - \lambda \delta_A - (1 - \lambda) \delta_R - r_\alpha > C_{n,\lambda}^{\text{stream}} + \delta_{\text{null}}. \quad (11)$$

Under the stated independent fixed-sample model, the false-positive probability is at most  $\alpha$ . Choosing  $n$  or  $\lambda$  after seeing data, optional stopping, or reporting the best of several lengths requires corrections not supplied by (11).

For an alternative score  $S_{\text{alt}}$ , define

$$g = S_{\text{alt}} - \lambda \delta_A - (1 - \lambda) \delta_R - C_{n,\lambda}^{\text{stream}} - \delta_{\text{null}}.$$

If  $g \leq 0$ , no amount of data certifies the claim under those allowances. If  $g > 0$ , fixed sample sizes satisfying  $r_\alpha + r_\beta < g$  guarantee power at least  $1 - \beta$  by a second Hoeffding bound. The implementation allocates trials asymptotically as  $N_A : N_R = \lambda : (1 - \lambda)$ .

## 5.2 Commitment and branch timing

A valid trial executes in this order:

1. prepare and independently characterise  $n$  EPR sources;
2. release  $A_i$  only after  $B_{i-1}$  has entered trusted sequestration;
3. after  $B_n$ , require delivery of the terminal memory port  $M$ ;
4. generate the hidden AUDIT/RETURN choice;
5. apply the preregistered branch decoder and binary acceptance measurement;
6. record every loss and exclusion under a predeclared rule.

The branch choice may be randomised per trial, but the final branch counts and stopping rule must be fixed or analysed with a sequentially valid method.

## 5.3 Mandatory shortcut controls

1. **Dephased memory:** deliberately dephase the persistent carrier between slots.
2. **Strong classical record:** measure every event and retain its transcript; AUDIT should improve while global RETURN falls.
3. **Collective access:** remove sequestration and verify the higher curve (9).
4. **Output revisit:** return an earlier  $B_i$  to the device, demonstrating why the causal boundary matters.
5. **Decoder null:** replace the coherent inverse with identity or a mismatched inverse.
6. **Source and measurement calibration:** bound false EPR acceptance and parity bias independently.
7. **Blind branch choice:** test for branch information leaking before commitment.
8. **Side-channel audit:** bound coherent transfer through buses, pulse tails, clocks, resonators, unused levels, and shared ancillas.

The null does not forbid inaccessible erasure environments. Consequently, visible reset of  $M$  is a reproducibility and wiring check, not a certificate that every physical copy of a classical transcript has vanished.

## 5.4 Reproducible analysis interface

The public implementation exposes three strict-JSON commands:

```
audit-return-benchmark bound --steps 8
audit-return-benchmark plan --steps 8 --forecast-model
audit-return-benchmark analyse --steps 8 \
  --audit-successes 9700 --audit-trials 10000 \
  --return-successes 9500 --return-trials 10000
```

Systematic allowances,  $\alpha$ ,  $\beta$ , and null slack are command-line inputs and appear in the output. An example preregistration records the terminal port, source calibration, decoder, exclusions, controls, and stopping rule.

## 6 Numerical validation and forecast

The theorem is analytic. Computation is used to attack the derivation, expose transitions, and convert a declared noise law into a shot budget; it is not used as a substitute for the proof.

## 6.1 Adversarial theorem checks

Independent scripts checked:

- the one- and two-slot closed forms and transition equations through  $n = 12$ ;
- random product likelihoods against the flagged recovery inequality;
- random collective likelihood tables against (9);
- the exact coherent-accumulator circuit;
- asymmetric local strengths and fully adaptive outcome trees for  $n = 2, \dots, 5$ .

The maximum numerical excess over the streaming theorem was  $6.52 \times 10^{-14}$  for adaptive trees and  $3.53 \times 10^{-14}$  for asymmetric products, consistent with numerical precision. The smallest tested collective-classical gap was positive. These checks do not prove the theorem or priority; they are regression tests against algebraic and implementation errors.

The release suite contains 25 automated tests. It verifies exact supports, transition behaviour, conservative systematic handling, fixed-sample planning, strict JSON output, deterministic simulation, and byte-stable same-environment figures and data.

## 6.2 Transparent forecast model

For planning only, we use the replaceable phenomenological law

$$P_A(n) = \frac{1 + c_0 c^n}{2}, \quad F_R(n) = f_0 f^n. \quad (12)$$

The committed defaults are  $c_0 = f_0 = 0.995$ ,  $c = 0.998$ , and  $f = 0.997$ . They are illustrative logical-operation factors, not hardware calibration.

With  $\lambda = 1/2$ ,  $\alpha = 0.01$ , target power 0.9, 0.005 systematic allowance in each branch, and 0.005 null slack, the model's robust plan remains feasible through  $n = 143$  and becomes infeasible at  $n = 144$ . The unadjusted expected score first crosses the 3/4 asymptotic streaming null at  $n = 151$ . Near the robust boundary the conservative Hoeffding shot count diverges rapidly, making uncertainty accounting more important than small circuit optimisations.

At  $n = 8$ , a declared alternative  $(P_A, F_R) = (0.97, 0.95)$  under the same allowances needs 84 trials per branch for the Hoeffding guarantee ( $\alpha = 0.01$ , power at least 0.9). If 9700 of 10000 AUDIT trials and 9500 of 10000 RETURN trials succeed, the adjusted one-sided lower score is approximately 0.9443, exceeding the enlarged 0.755 null by approximately 0.1893.

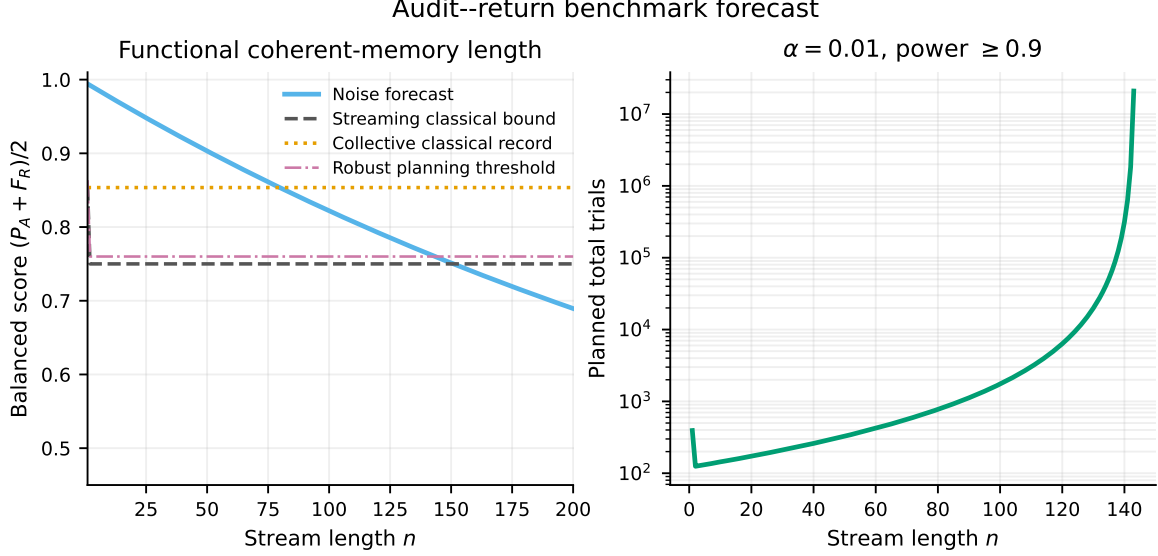


Figure 2: Illustrative forecast from (12). Left: balanced score against the streaming, collective-classical, and robust planning thresholds. The one-slot streaming point is higher than the  $n \geq 2$  plateau. Right: conservative total trial count from (10); infeasible lengths are omitted. These curves are not hardware predictions.

### 6.3 Functional coherent-memory length

We define the *certifiable functional coherent-memory length* as the largest preregistered  $n$  for which (11) is satisfied under a declared familywise testing rule. Unlike a bare  $T_2$ , it asks a memory to retain a temporal predicate and later participate in coherent global return. It remains benchmark-specific: changing the predicate, branch scores, or sequestration interface changes the resource being measured.

## 7 Experimental feasibility

### 7.1 Minimal logical circuit

The honest coherent device needs one persistent qubit and  $n$  streamed carrier interactions. In the ideal binary implementation:

1. the source creates  $n$  EPR pairs;
2. the device applies one  $\text{CNOT}_{A_i \rightarrow M}$  per slot;
3. a trusted switch routes every returned  $B_i$  to storage inaccessible to the device;
4. after the terminal memory is committed, AUDIT measures  $M$  and the  $R_i$  in  $Z$ ;
5. RETURN applies  $n$  inverse CNOTs from  $B_i$  to  $M$ , then Bell-tests every  $R_i B_i$  and tests  $M = 0$ .

The logical depth inside the untrusted device is one two-qubit gate per slot. The recovery station has  $n$  further two-qubit gates and  $n$  Bell analysers. The global RETURN event is deliberately stringent: every pair and the terminal reset must pass in the same trial.

### 7.2 The hard part is the interface

Gate counts are modest; causal enforcement is not. An implementation needs all of:

- a switch or transport geometry that proves  $B_i$  left the device before  $A_{i+1}$  arrived;
- storage of every  $B_i$  with sufficiently low loss and dephasing for the later joint decoder;

- a terminal port that transfers  $M$  without revealing the branch choice;
- calibrated suppression of coherent paths that cross slot boundaries outside the charged memory;
- a random branch generator and control electronics whose timing cannot leak into the device.

A hidden bus mode or coherent pulse tail is not “noise” under the theorem: it is precisely a quantum-memory resource that moves the apparatus outside the null. Such a side channel may still produce a valid positive witness of some coherent memory, but it prevents attributing the result to the nominated memory qubit.

### 7.3 Candidate platforms

Superconducting circuits can combine fast controlled gates, a dedicated memory or bus mode, and microwave routing, but resonator and control-line remnants require an explicit side-channel budget. Trapped ions offer long-lived internal memories and high-quality entangling gates, while physical sequestration is naturally expressed through shuttling, zones, or trusted spectator isolation. Neutral-atom arrays offer reconfigurable geometry and parallel entanglement tests, with loss treatment needing preregistration. Photonic time bins coupled to a matter memory naturally realise a stream, but source loss, storage loss, and postselection are central risks.

The protocol does not select a platform by headline fidelity alone. The decisive platform metric is whether the access model can be made credible and independently audited.

### 7.4 Staged demonstration

1.  **$n = 2$  logical demonstration:** verify the coherent point and the dephased-memory control.
2. **Classical frontier:** implement weak diagonal instruments and recover the predicted support curve as  $\lambda$  varies.
3. **Causal separation:** compare enforced streaming with the collective-access control on the same hardware.
4. **Length scaling:** preregister  $n$  values and report simultaneous confidence intervals or corrected tests.
5. **Independent interface audit:** characterise leakage channels without using the primary score data.

### 7.5 Kill criteria

The flagship claim must be reduced or abandoned if:

1. the theorem’s classical-path factorisation does not match the stated experimental null;
2. an adaptive classical strategy exceeds (5);
3. the branch choice or sequestered carriers become accessible before commitment;
4. postselection or loss rules can condition the score on hidden outcomes;
5. source correlations invalidate independence of the  $X_i$ ;
6. the reported score uses average pair fidelity instead of the global RETURN event;
7. an earlier theorem is found that directly specialises to the full interface and support;
8. systematic allowances needed to defend the access model erase the statistical gap.

Any of these outcomes would still leave useful calibration or pedagogy, but not the claimed exact resource witness.

## 8 Discussion

The benchmark is useful because it turns an imprecise question about reversible records into one number with an exact null. A high AUDIT score alone may come from destructive classical measurement. A high RETURN score alone may come from doing nothing. Requiring both after a hidden late choice makes persistent coherence useful, while immediate sequestration prevents a classical device from compiling the global parity into one collective measurement.

The first-order transition is a structural consequence of asking for a parity of many fresh events. Classical local evidence combines multiplicatively: a modest per-slot record becomes exponentially weak as a parity predictor. Global return fidelity also multiplies. For  $n \geq 3$ , the supported optimum therefore avoids intermediate strategies over a range of weights and jumps from no measurement to almost projective measurement. The phenomenon is a decision transition in the optimal null, not a thermodynamic phase transition in matter.

The strict gap between streamed and collective classical recording is equally important. It demonstrates why a temporal access model can reveal a resource separation that a one-shot information–disturbance curve cannot. The coherent accumulator then shows that the task does not demand a growing quantum memory: one protected temporal qubit is sufficient for an arbitrarily long parity stream in the noiseless circuit. Experimentally, of course, global return becomes increasingly fragile with  $n$ .

### 8.1 Limitations

The benchmark is not device independent. It trusts the EPR source, the branch measurements, the sequestration station, the timing boundary, and the absence or calibrated allowance of uncharged coherent side channels. It certifies incompatibility with a declared classical-memory comb; it does not reconstruct the device’s internal dynamics or prove that the named qubit was the only memory.

Visible terminal reset is not global erasure. A classical controller may reset its public port while dumping its transcript into an inaccessible environment. The present null allows this and the theorem remains valid. Demanding decoupling from every purification would define a different, stronger task.

The finite-statistics result assumes independent Bernoulli trials and fixed sample sizes. Drift, correlated errors, adaptive stopping, multiple lengths, and estimated systematic bounds need dedicated inference. The phenomenological forecast is a planning example, not a hardware claim.

No interpretation of quantum mechanics is selected. A successful result says that a coherent temporal resource outperforms an explicit classical-memory model on an ordinary quantum-information task.

### 8.2 Extensions

Natural next problems are:

1. exact frontiers for  $q$ -ary streams and predicates beyond parity;
2. tight bounds for a finite coherent memory dimension between zero and one full logical qubit per task;
3. robust nulls with bounded cross-slot leakage rather than a sharp classical/quantum dichotomy;
4. sequentially valid inference for random stopping and adaptive length selection;
5. semi-device-independent or device-independent variants with operationally certified sequestration;
6. hardware experiments that compare streamed and collective access using the same physical components.

### 8.3 Conclusion

For a specific streamed parity task, unlimited adaptive classical memory obeys an exact audit–return frontier, collective classical processing performs strictly better, and one coherent temporal qubit reaches the algebraic optimum. This is a compact, falsifiable target for quantum-memory experiments. The main theoretical work is complete under the declared null; the main experimental work is to make that null physically credible.

## A Proofs

### A.1 Flagged entanglement-recovery lemma

**Lemma A.1.** *Let  $\mathcal{N}_k : A \rightarrow B$  be a trace-nonincreasing CP map,  $\dim A = D$ , and*

$$p(k|x) = \text{Tr } \mathcal{N}_k(|x\rangle\langle x|)$$

*in a fixed orthonormal basis. For any recovery channel  $\mathcal{R}_k : B \rightarrow A$ , its unnormalised contribution to maximally entangled fidelity satisfies*

$$F_k \leq \frac{1}{D^2} \left( \sum_x \sqrt{p(k|x)} \right)^2. \quad (13)$$

*The bound is attainable for every valid likelihood table by a positive diagonal instrument.*

*Proof.* Put  $\Lambda_k = \mathcal{R}_k \circ \mathcal{N}_k$ . Expanding against  $|\Phi_D\rangle = D^{-1/2} \sum_x |x\rangle_R |x\rangle_A$  gives

$$D^2 F_k = \sum_{x,y} \langle x | \Lambda_k(|x\rangle\langle y|) | y \rangle.$$

Positivity of the Choi matrix implies

$$|\langle x | \Lambda_k(|x\rangle\langle y|) | y \rangle| \leq \sqrt{\langle x | \Lambda_k(|x\rangle\langle x|) | x \rangle \langle y | \Lambda_k(|y\rangle\langle y|) | y \rangle} \leq \sqrt{p(k|x)p(k|y)}.$$

Summing proves (13). For tightness, choose  $K_k = \sum_x \sqrt{p(k|x)} |x\rangle\langle x|$  and identity recovery. Completeness of the likelihood table makes  $\sum_k K_k^\dagger K_k = \mathbb{I}$ .  $\square$

### A.2 Causal tensorisation

For a refined transcript  $k$  and the definitions in (2), every fresh  $X_i$  is uniform and independent of the previous transcript. Bayes' rule and (1) therefore make the posterior distribution of  $X_1, \dots, X_n$  a product along the realised leaf. The conditional parity bias is

$$\left| \mathbb{E}[(-1)^{X_1 + \dots + X_n} | k] \right| = \prod_i \delta_i.$$

Averaging the optimal parity guess gives

$$P_A = \frac{1}{2} \left( 1 + \mathbb{E}_k \prod_i \delta_i \right). \quad (14)$$

Apply (13) with  $D = 2^n$ , use (1), and sum the input strings independently at every slot:

$$F_R \leq 4^{-n} \sum_k \left( \sum_x \sqrt{p(k|x)} \right)^2 \quad (15)$$

$$= \sum_k \prod_i \frac{\left( \sqrt{p_i(k_i|0, h)} + \sqrt{p_i(k_i|1, h)} \right)^2}{4} \quad (16)$$

$$= \mathbb{E}_k \prod_i \frac{1 + \sqrt{1 - \delta_i^2}}{2} = \mathbb{E}_k \prod_i f(\delta_i). \quad (17)$$

Thus

$$S_{n,\lambda} \leq \frac{\lambda}{2} + \mathbb{E}_k \left[ \frac{\lambda}{2} \prod_i \delta_i + (1 - \lambda) \prod_i f(\delta_i) \right].$$

An expectation cannot exceed the largest realised bracket, proving the upper bound in (4). The diagonal binary family in Section 3 attains every deterministic strength profile, proving tightness and showing that adaptive or non-QND freedom cannot improve the support.

### A.3 Equal strengths

Set

$$t_i = \frac{2y_i}{1 + y_i^2}, \quad f(t_i) = \frac{1}{1 + y_i^2}, \quad 0 \leq y_i \leq 1.$$

Apart from the additive  $\lambda/2$ , the objective is

$$G(y_1, \dots, y_n) = \frac{(1 - \lambda) + \lambda 2^{n-1} \prod_i y_i}{\prod_i (1 + y_i^2)}.$$

At fixed nonzero  $Y = \prod_i y_i$ , let  $z_i = \log y_i$ . Since  $\phi(z) = \log(1 + e^{2z})$  is strictly convex, Jensen's inequality minimises the denominator at  $z_1 = \dots = z_n$ , hence  $y_1 = \dots = y_n$ . If  $Y = 0$ , setting every  $y_i = 0$  minimises the denominator. This proves (5) globally, including boundaries.

### A.4 Closed forms and first-order transition

For common  $y$ , the nonconstant part of the support is

$$g_n(y) = \frac{1 - \lambda + \lambda 2^{n-1} y^n}{(1 + y^2)^n}.$$

Direct maximisation gives the one-slot circle and (6). For  $n \geq 3$ ,

$$g'_n(y) = \frac{ny}{(1 + y^2)^{n+1}} \left[ \lambda 2^{n-1} y^{n-2} (1 - y^2) - 2(1 - \lambda) \right].$$

The endpoint  $y = 0$  remains locally maximal. Requiring a nonzero stationary point to tie it eliminates  $\lambda$  and yields (7); substituting back gives (8). The solution interval in (7) contains exactly one root because the left side decreases from a value above one to zero after its maximum at  $(n - 2)/n$ . Since the tying  $y$  is nonzero, the exposed optimal strength jumps. Expanding around  $z_n = 1$  gives

$$1 - z_n = 2^{1-n} + \frac{n-1}{2} 2^{2-2n} + O(n^2 2^{3-3n}), \quad \lambda_c(n) = \frac{2}{3} - \frac{2^{1-n}}{9} + O(n 2^{2-2n}).$$

### A.5 Collective classical-record optimum

Let  $D = 2^n$  and  $d = D/2$ . For a collective outcome  $k$ , set

$$a_k = \sum_{x: \text{par}(x)=0} p(k|x), \quad b_k = \sum_{x: \text{par}(x)=1} p(k|x).$$

Optimal parity decoding gives

$$P_A = \frac{1}{2} + \frac{1}{2D} \sum_k |a_k - b_k|.$$



The recovery lemma followed by Cauchy–Schwarz within each parity sector gives

$$F_R \leq \frac{1}{D^2} \sum_k (\sqrt{da_k} + \sqrt{db_k})^2.$$

With  $A_k = a_k/d$ ,  $B_k = b_k/d$ ,

$$T = \frac{1}{2} \|A - B\|_1, \quad C = \sum_k \sqrt{A_k B_k},$$

we have  $P_A = (1 + T)/2$  and  $F_R \leq (1 + C)/2$ . The classical fidelity–total-variation relation  $T^2 + C^2 \leq 1$  yields  $F_R \leq f(T)$ . Optimising the resulting one-variable score proves (9). A weak measurement of the two parity projectors attains equality.

Finally, for any  $n \geq 2$ ,  $\prod_i f(t_i) \leq f(\prod_i t_i)$ . The collective device can therefore reproduce the same parity bias with no greater disturbance and is strictly better in every interior support direction except at trivial equality cases. This proves the strict resource gap.

## B Reproducibility specification

The release artifact uses Python 3.10.5, NumPy 2.2.6, SciPy 1.15.3, Matplotlib 3.10.9, and pytest 9.1.1. From the repository root:

```
python -m pip install -e ".[dev,reproducibility]"
python -m pytest -q
python scripts/regenerate.py --seed 20260812 --shots 8192
python scratch/originality_gate/validate_streaming_parity.py
python scripts/build_pdf.py
```

The committed audit–return outputs are:

- `data/audit_return_frontier.csv`: 606 exact support rows for six stream lengths and 101 weights;
- `data/audit_return_forecast.csv`: 200 planning rows under the declared example law;
- `figures/audit_return_frontier.pdf`: exact support and strategy-transition figure;
- `figures/audit_return_benchmark.pdf`: functional-length and power forecast;
- `scratch/originality_gate/streaming_parity.json`: stored expensive adversarial searches;
- `streaming_parity_validation.json`: independent fast validation summary.

PDF timestamps are suppressed for deterministic same-environment generation. Raster and vector rendering can differ across operating systems because of font and rendering-library versions; numerical CSV drift is the stronger cross-platform invariant.

The forecast does not simulate a specific device’s density matrix. It evaluates the exact theorem and statistical planner under (12). Experimental use must replace its probabilities with measured or independently calibrated values.

## C Claim ledger

The exact allowed novelty sentence is:

To our knowledge, for the precisely specified streaming class, we derive the exact parity-audit versus global-entanglement-return support, including arbitrary adaptive non-QND instruments, and prove a strict same-task separation from collective classical-record strategies and from a one-qubit coherent accumulator.

The sentence must be revised if priority review identifies a direct antecedent.

Table 2: Permitted claims and their evidence status at the 12 August 2026 cutoff.

| Claim                                                                 | Status               | Boundary                                                                    |
|-----------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------|
| Equation (5) is the exact adaptive classical-memory support           | Proved here          | Requires the factorised classical-path null and trusted access model        |
| The balanced null is 3/4 for $n \geq 2$                               | Corollary            | Uses global RETURN fidelity                                                 |
| The $n \geq 3$ exposed optimum has a first-order transition           | Proved here          | A decision transition of the optimiser, not a material phase transition     |
| Collective classical recording obeys (9)                              | Proved here          | “Collective” does not mean coherent memory is forbidden unless stated       |
| One persistent coherent qubit attains unit score                      | Exact circuit        | Assumes ideal gates, source, storage, and decoder                           |
| The fixed-sample rule has false-positive probability at most $\alpha$ | Proved from Hoffding | Independent Bernoulli trials, preregistered fixed sample and weight         |
| Defaults remain robustly planifiable through $n = 143$                | Numerical forecast   | Illustrative model, not hardware calibration                                |
| No exact prior theorem was located                                    | Dated search result  | Not proof of priority; subject to specialist review                         |
| A positive experiment certifies coherent temporal memory              | Model-relative       | Rejects the declared null, not every classical or hidden-device explanation |
| The protocol deletes every record or supports an interpretation       | Not claimed          | Neither follows from the score                                              |

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